Experiment Number : 10 **Date: 27-02-26**

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24BCS285

CSE-A1

Implementation of Access Modifiers in Java

Aim:

Write a Java program to implement different access modifiers.

PRE LAB EXERCISE

QUESTIONS

1. What are the 4 access modifiers available in Java?

- **private**
- **default** (no keyword)
- **protected**
- **public**

2. Which access modifiers can be used in Java to control access to class members?

- **private**
- **default**
- **protected**
- **public**

(All four can be used for class members like variables and methods.)

IN LAB EXERCISE

Objective

To demonstrate different access modifiers such as Default, Private, Protected and Public using Java programs.

	Default	Private	Protected	Public
Same Class	Yes	Yes	Yes	Yes
Same Package Subclass	Yes	No	Yes	Yes
Same Package Non-Subclass	Yes	No	Yes	Yes
Different Package Subclass	No	No	Yes	Yes
Different Package Non-Subclass	No	No	No	Yes

Fig: Comparison table of Access Modifiers in Java

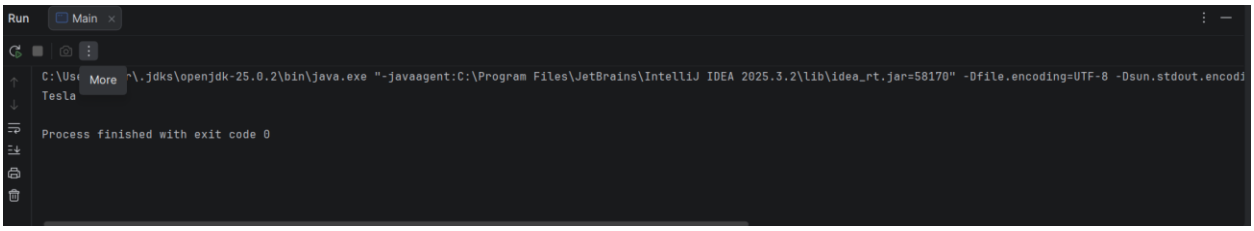
Source Code

// Default Access modifier

```
class Car {
   String model; // default access
}
public class Main {
      public static void main(String[] args){
       Car c = new Car();
       c.model = "Tesla"; // accessible within the same package
       System.out.println(c.model);
   }
}
```

Output:

Tesla



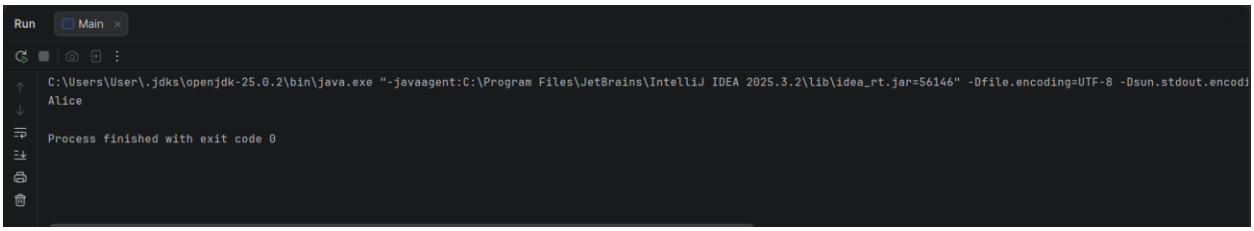
//Private Access Modifier

```
class Person {
    // private variable
    private String name;
    public void setName(String name)  {
     this.name = name; // accessible within class
    }
    public String getName() {
            return name;
   }
}
public class Geeks {
    public static void main(String[] args)
    {
       Person p = new Person();
        p.setName("Alice");

        // System.out.println(p.name);
        // Error: 'name'
        // has private access
        System.out.println(p.getName());
    }
}
```

Output

Alice



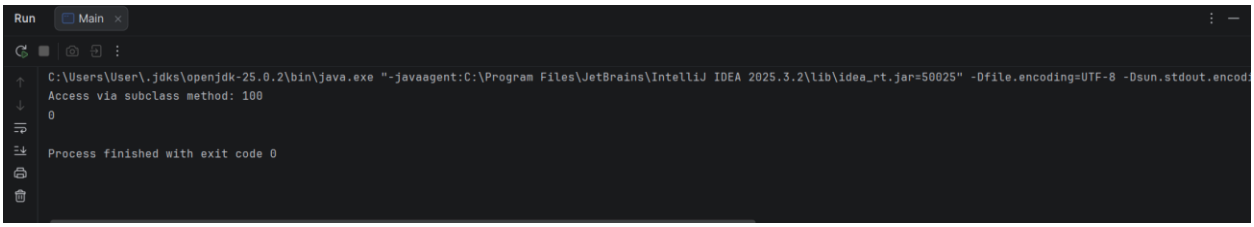
//Protected Access Modifier

```
class Vehicle {
    protected int speed; // protected member
}
class Bike extends Vehicle {
    void setSpeed(int s)
    {
        speed = s; // accessible in subclass
    }
    int getSpeed()
    {
        return speed; // accessible in subclass
    }
}
public class Main {
    public static void main(String[] args){
        Bike b = new Bike();
        b.setSpeed(100);
        System.out.println("Access via subclass method: "+ b.getSpeed());
        Vehicle v = new Vehicle();
        System.out.println(v.speed);
    }
}
```

Output

Access via subclass method: 100

0

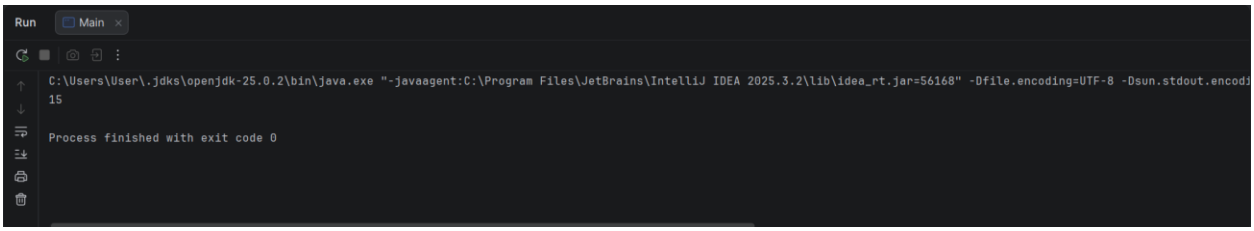


//Public Access Modifier

```
class MathUtils {
        public static int add(int a, int b) {
        return a + b;
    }
}
public class Main {
        public static void main(String[] args) {
        System.out.println(MathUtils.add(5, 10)); // accessible anywhere
    }
}
```

Output

15



POST LAB EXERCISE

1. Write a Java program to implement Default access modifier which has a default class within the same package and a default class from a different package.

Same Package (Works)

Java

```java
// package1/DefaultClass.java
package package1;

class DefaultClass {
    void show() {
        System.out.println("Inside same package");
    }
}
```

Java

```java
// package1/Test.java
package package1;

public class Test {
    public static void main(String[] args) {
        DefaultClass obj = new DefaultClass(); // Works
        obj.show();
    }
}
```

Different Package (Not Works)

Java

```java
// package2/Test2.java
package package2;

import package1.DefaultClass; // Error

public class Test2 {
    public static void main(String[] args) {
        DefaultClass obj = new DefaultClass(); // Not allowed
    }
}
```

Default class is accessible **only within same package**.

2. Which access modifier provides the highest level of access?

public

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	